

Extreme Events Attribution

Weekly meeting

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Section 1

Detection and attribution of climate extremes in the observed record

Classical Detection and Attribution Framework

Detection

- Determine whether an observed change is distinguishable from internal climate variability.

Attribution

- Evaluate the contributions of potential causes with statistical confidence.
- Anthropogenic forcings (ANT): greenhouse gases, aerosols, ozone, and land-use change.
- Natural forcings (NAT): solar variability and volcanic activity.

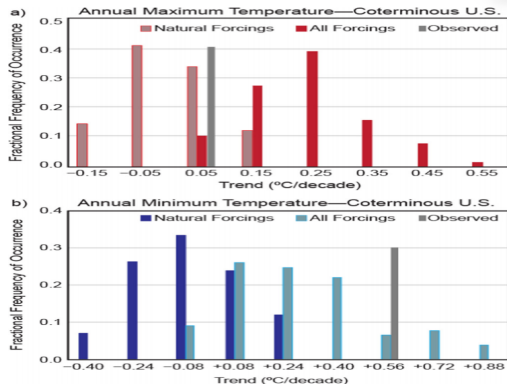
Model-based comparison

- 1 Identify a spatial or temporal signal in observations.
- 2 Compare it with **historical simulations** containing natural and anthropogenic forcings.
- 3 Compare it with **natural-only simulations**, representing a counterfactual climate.
- 4 Attribute the signal to human influence if it is reproduced by historical simulations but not by natural-only simulations.

Example: U.S. Extreme Temperature Trends

Study design

- Period: 1956–2005
- Data source: GHCND (Gridded observational data)



CMIP5 simulations

- Historical: 78 simulations from 29 models
- Natural-only: 35 simulations from 15 models

Reading Figure 6

- Red: distribution of historical trends
- Blue: distribution of natural-only trends
- Gray: observed U.S. trend
- T_{xx} : $+0.01^{\circ}\text{C}/\text{decade}$; not significant and within both distributions
- T_{nn} : $+0.56^{\circ}\text{C}/\text{decade}$; significant and outside the natural-only distribution

Examples of Trend Attribution Studies

Study	Response	Predictors / signals	Data	Method
Min et al. (2011)	Changes in annual maximum 1-day and 5-day precipitation	Anthropogenic and natural forcing fingerprints	HadEX observations, 1951–2003; CMIP3 20c3m, ant, and nat simulations	Linear PCA-based optimal fingerprinting
Min et al. (2013)	5- and 10-year means of T_{xx} , T_{nx} , T_{xn} , and T_{nn}	Anthropogenic and natural forcing fingerprints	Observed extreme-temperature indices, 1951–2000; CMIP3 multi-model ensemble	Optimal fingerprinting at global and five continental scales
Christidis et al. (2013)	Changes in daily extreme temperatures, including warm days and nights	GHG forcing and land-use change	Observed extreme-temperature indices and climate-model experiments	Factorial attribution separating GHG and land-use effects

Probabilistic Extreme Event Attribution

Research question

- “If climate change directly cause this specific event?” **TO** “How did anthropogenic climate change alter the probability or magnitude of this event?”

Probability ratio

$$PR = \frac{P_{\text{real}}}{P_{\text{nat}}}$$

- P_{real} : event probability in the factual world, ie, $P(\text{event} \mid \text{NAT} + \text{ANT})$.
- P_{nat} : event probability in a counterfactual natural world, ie, $P(\text{event} \mid \text{NAT only})$
- $PR > 1$: anthropogenic climate change made the event more likely. For example, $PR = 2$ means that the event became twice as likely

Fraction of attributable risk

$$FAR = 1 - \frac{P_{\text{nat}}}{P_{\text{real}}} = 1 - \frac{1}{PR}$$

- FAR measures the fraction of the current event **risk** associated with anthropogenic influence.

Two Approaches to Event Attribution

	Oxford approach	Boulder approach
Primary question	How much more likely did climate change make the event?	How much did climate change alter the magnitude of the event?
Main comparison	Factual versus counterfactual model ensembles	Natural and anthropogenic contributions to the event's meteorology
Main quantities	PR & FAR	Changes in intensity or magnitude
Examples	2003 European heat wave: Stott et al. (2004) estimated that anthropogenic influence at least doubled its probability	2010 Russian heat wave: Dole et al. (2011); 2011 Texas heat wave and drought: Hoerling et al. (2013)

Connection between the two approaches

A small anthropogenic change in magnitude can produce a large change in probability when an event lies in the tail of the distribution.

Sources of Uncertainty in Event Attribution

① Observational uncertainty

- The observed magnitude of an event may be uncertain.
- This uncertainty can affect the estimated probability ratio.

② Counterfactual uncertainty

- The natural world without anthropogenic influence cannot be observed.
- Counterfactual sea-surface temperatures and sea ice must be estimated.
- These estimates depend on assumptions about climate sensitivity.

③ Model structural uncertainty

- Different climate models can produce different extreme-weather responses under the same boundary-condition changes.

④ Conditionality

- Attribution conclusions are conditional on the selected model, forcing assumptions, boundary conditions, and event definition.

Section 2

Statistical Methods for Extreme Event Attribution in Climate Science

Statistical Models for Extreme Event Attribution

Section 5 (Statistical Methods) reviews four major statistical approaches used to estimate event probabilities in the factual and counterfactual worlds:

- ① Nonparametric approaches
- ② Univariate extreme value theory (EVT)
- ③ Hierarchical modeling
- ④ Multivariate extreme value theory

The main objective is to estimate

$$p_0 = P_{\text{nat}} \quad \text{and} \quad p_1 = P_{\text{all}},$$

and subsequently quantify the anthropogenic influence through

$$RR = PR = \frac{p_1}{p_0}, \quad FAR = 1 - \frac{p_0}{p_1}.$$

Nonparametric Approaches

The simplest approach directly counts threshold exceedances in the factual and counterfactual simulation ensembles.

For an event defined by $Y > u$, the FAR can be estimated by

$$\widehat{\text{FAR}}_n(u) = 1 - \frac{\sum_{i=1}^n I(X_i > u)}{\sum_{i=1}^n I(Z_i > u) + 1/2},$$

where

- X_i : counterfactual or natural-forcing simulations;
- Z_i : factual or all-forcing simulations;
- $1/2$: correction used to avoid division by zero.

Main advantage: No parametric distributional assumption is required.

Main limitation: The method requires very large ensembles when the event is rare.

Limitations of Nonparametric Estimation

For a fixed threshold u , the estimator is asymptotically Gaussian:

$$\sqrt{n} \left[\widehat{\text{FAR}}_n(u) - \text{FAR}(u) \right] \xrightarrow{d} N(0, \sigma^2(u)),$$

with

$$\sigma^2(u) = \frac{p_0^2}{p_1^2} \left(\frac{1 - p_0}{p_0} + \frac{1 - p_1}{p_1} \right).$$

- As the event becomes rarer, p_0 and p_1 become small.
- The estimator variance therefore increases rapidly.
- The ensemble size must be much larger than the event return period.
- Zero exceedances in the counterfactual ensemble make the risk ratio particularly difficult to estimate.

Large ensembles such as *weather@home* can reduce sampling uncertainty, but are computationally expensive and often estimate conditional rather than unconditional event probabilities.

Univariate Extreme Value Theory

Extreme value theory models the distribution tail and estimates probabilities beyond the observed or simulated range.

Generalized Extreme Value (GEV)

Used for block maxima, such as annual maximum temperature:

$$G(x, \mu, \alpha, \xi) = \exp \left\{ -\bar{H}_\xi \left(\frac{x - \mu}{\sigma} \right) \right\}.$$

- One maximum per time block
- Suitable for annual or seasonal maxima

The shape parameter ξ determines whether the tail is heavy, exponential-type, or bounded.

Generalized Pareto Distribution (GPD)

Used for exceedances above a high threshold v :

$$P(X > x \mid X > v) = \bar{H}_\xi \left(\frac{x - v}{\sigma} \right).$$

- Uses all threshold exceedances
- May use extreme observations more efficiently

Challenges in Applying EVT to Event Attribution

EVT is useful for very rare events, but it is not automatically appropriate for every event labeled as “extreme.”

① **The event may not be sufficiently extreme.**

EVT is most useful for estimating very high quantiles, such as the 0.99, 0.999, or 0.9999 quantiles.

② **Threshold selection is difficult.**

A low threshold gives more observations but may violate the GPD approximation; a high threshold reduces bias but increases variance.

③ **Available data have different structures.**

Observational records are often short, while model ensembles may be large but conditional on prescribed climate boundary conditions.

④ **Nonstationarity must be considered.**

Distribution parameters may change with time, forcing level, global mean temperature, or other climate covariates.

Hierarchical Modeling

Bayesian hierarchical models provide a framework for combining multiple sources of uncertainty:

- the unknown true climate response to external forcing;
- internal climate variability;
- differences among climate models;
- uncertainty in estimated covariance structures;
- observational and model errors.

Katzfuss et al. (2017) combined:

- an errors-in-variables model;
- principal component dimension reduction;
- Bayesian model averaging.

Hannart and Naveau (2018) extended this framework by selecting a linear projection that maximizes the evidence for a causal relationship.

Limitation: These models often rely on Gaussian assumptions and may not adequately represent non-Gaussian variables or extreme distribution tails.

Multivariate Extreme Value Theory

Many climate extremes involve multiple variables, locations, or time periods, such as simultaneous heat and drought.

The upper-tail dependence coefficient is defined as

$$\chi = \lim_{q \rightarrow 1} P[F_1(Y_1) > q \mid F_2(Y_2) > q].$$

- A larger χ indicates stronger dependence between joint extremes.
- Multivariate GPD models describe observations after at least one component exceeds an extreme threshold.
- Linear projections can reduce a multivariate extreme event to a univariate GPD model.
- Multivariate Pareto processes can model spatial or temporal extremes.

Ignoring dependence between variables or locations may substantially underestimate the probability of compound and systemic climate risks.

Asymptotic Dependence and Independence

Two variables may be strongly correlated at observed levels but become independent in the far tail.

- **Asymptotic dependence:** joint extremes remain dependent as the threshold increases.
- **Asymptotic independence:** the conditional probability of one variable being extreme approaches zero as the other becomes increasingly extreme.

The Ledford–Tawn coefficient η describes the rate of joint-tail decay:

$$P(Y_1 > y \mid Y_2 > y) \sim L(y)y^{1-1/\eta}.$$

$$\eta = 1 \quad \Rightarrow \quad \text{asymptotic dependence.}$$

Incorrectly assuming asymptotic dependence may substantially overestimate the probability of extremely rare compound events.

Comparison of Statistical Methods

Method	Main Idea	Best Used When	Main Limitation
Nonparametric	Directly count threshold exceedances	Large ensembles and moderately rare events	Large uncertainty for very rare events
Univariate EVT	Model and extrapolate the distribution tail	A single variable and probabilities beyond the sample range	Threshold selection and model assumptions
Hierarchical models	Combine multiple sources of uncertainty	Multiple models, observations, and uncertain climate responses	Often relies on Gaussian assumptions
Multivariate EVT	Model dependence among joint extremes	Compound, spatial, or temporal extreme events	Tail dependence is difficult to identify and estimate

Main conclusion: No single method currently integrates all sources of uncertainty, tail behavior, causal quantities, and multivariate dependence in extreme event attribution.

Section 3

Thoughts

Thoughts on our review paper

Base on the review paper I have read, we could have the structure Like:

- ① Introduction: Background
- ② Setup: including concept clarification, Denote.
- ③ Data: clarify current main data source and their character.
- ④ Methods for Classical Event Attribution.
- ⑤ Methods for Probabilistic Event Attribution.

Thanks

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Extreme Events Attribution